Contradiction Metabolism for LLMs: Preliminary Evidence that Context Rot is a Knowledge Integrity Problem

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Abstract

Large Language Models degrade during extended conversations—a phenomenon known as "context rot." The prevailing assumption is that this results from context length limitations. We present preliminary evidence suggesting that *contradiction accumulation*, not context length, is a primary driver of this degradation.

In a controlled three-condition experiment with gemma3:27b ($n = 3$ per condition, 180 turns), accuracy under contradiction injection was 21.1%, compared with 73.3% when an external metabolism layer resolved contradictions during idle time, and 56.7% in a contradiction-free baseline. All pairwise differences reach the minimum achievable Mann-Whitney $p = 0.05$ with $n = 3$; Kruskal-Wallis $p = 0.027$. A cross-model sign test across 8 open-source models (11 trial pairs) shows 9 of 10 non-tied pairs favoring metabolism ON ($p = 0.0107$); a more conservative by-model analysis (6 of 7 models, $p = 0.0625$) is consistent but borderline.

Separately, single-run experiments across 6 models from 4 vendors suggest that contradiction accumulation impairs accuracy regardless of context window size (8K–1M tokens), with each model tested once ($n = 1$). Frontier model replication suggests a model-specific collapse threshold (δ_c): lightweight models collapse catastrophically, while frontier models show conditional resistance. A factor analysis ($n = 1$ per condition) further suggests a capability–vulnerability paradox in which high-capability models comply with coercive contradiction injections that lower-capability models reject. A lightweight middleware (`delta-prune`) reduced this compliance in a single preliminary test ($n = 1$).

While most metabolism experiments use open-source models (8B–27B parameters), completed frontier full-pipeline replications on Sonnet 4.6 and Gemini 3.1 Flash Lite ($n = 3$, 30 turns) now show the same qualitative ordering: `ON` $\approx$ `NC` and `ON` $\gg$ `OFF`. On Sonnet, mean T30 fact recall is 95.6% (`ON`), 97.8% (`NC`), and 46.7% (`OFF`); on Gemini, the corresponding means are 80.0%, 88.9%, and 2.2%. An audited GPT-4o line provides supportive but quarantine-limited secondary evidence. Larger samples, longer-horizon frontier validation, and cross-language validation are needed to establish generality.

1 Introduction

1.1 The Context Rot Problem

Large Language Models increasingly serve as long-running conversational agents—personal assistants, coding partners, and knowledge workers that maintain extended dialogue sessions spanning hundreds of turns. In such settings, a well-documented but poorly understood phenomenon occurs: the model's ability to recall facts, apply rules, and maintain coherent reasoning degrades over the course of the conversation. We refer to this degradation as *context rot*.

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The prevailing industry response treats context rot as a length problem. Google introduced Gemini models with 1M-token context windows [Gemini Team, 2025]. Anthropic expanded Claude's context to 200K tokens. OpenAI added persistent Memory to ChatGPT. The implicit assumption is that if the context window is large enough, degradation will not occur.

1.2 Not Length, But Contradictions

Our experiments tell a different story. Table 1 presents results from a controlled experiment across 6 models from 4 vendors. Each model was tested under two conditions: conversations with no contradictions ($\delta = 0$) and conversations with contradictory information injected at regular intervals ($\delta > 0$). The same conversation length was used in both conditions.

Table 1: Effect of contradiction accumulation ($\delta > 0$) on LLM accuracy across models and context window sizes. Data from Experiment 35 in Sunagawa [2026], using a controlled dialogue protocol distinct from the DeltaZero experiments in Sections 4.2–4.4. Each cell is a single run ($n = 1$).

Model	Vendor	Context	$\delta = 0$	$\delta > 0$	Drop
GPT-4o-mini	OpenAI	96K	100%	10.4%	−89.6pp
Gemini 2.5 Flash	Google	64K	100%	0%	−100.0pp
Gemini 3.1 FL	Google	1M	88.6%	40.8%	−47.8pp
Sonnet 4	Anthropic	8K	100%	74.0%	−26.0pp
Sonnet 4.6	Anthropic	128K	100%	100%*	0pp
Llama 3.1:8b	Meta	8K	34.0%	2.7%	−31.3pp

*Recognition accuracy. Strict output parsing yields 82.6% due to format variability.

The critical comparison is between context window size and degradation magnitude. Sonnet 4, with an 8K context window, drops 26.0 percentage points (pp) under contradiction. Gemini 3.1 FL, with a 125× larger 1M-token window, drops 47.8pp—*worse*, not better. Yet when contradictions are removed ($\delta = 0$), all models maintain accuracy regardless of context length. The causal variable is contradiction density, not token count.

Figure 1 visualizes this contrast. Gemini 2.5 Flash collapses completely (−100pp), while the $\delta = 0$ condition shows near-perfect performance across models.

One model stands out: Sonnet 4.6 shows zero degradation under our contradiction protocol. A clarification on the 100% figure: when evaluated on whether the model *recognizes* that contradictory information has been presented, Sonnet 4.6 identifies all contradictions correctly. The lower strict-parsing score (82.6%) reflects output format variability—the model communicates the correct answer but not always in the expected keyword format. This is a measurement artifact, not a model failure; we report the recognition accuracy as the more meaningful metric. However, Sonnet 4.6 *tolerates* contradictions rather than managing them—it identifies conflicts when asked but does not resolve, organize, or prioritize conflicting information. For the remaining 5 of 6 models tested, collapse under contradiction is severe and universal.

1.3 Cognitive Sleep: A Biological Analogy

Humans face a similar challenge. During waking hours, we accumulate new information that may conflict with existing beliefs. The brain does not resolve these conflicts in real time; instead, it consolidates memory during sleep—organizing facts, resolving inconsistencies, and forgetting irrelevant details [Walker, 2017, Stickgold, 2005]. Sleep deprivation, which prevents this consolidation, leads to progressive cognitive degradation remarkably similar to context rot in LLMs.

Current LLMs have no equivalent of sleep. They accumulate all information—consistent and contradictory alike—without any mechanism for offline processing. We draw an analogy

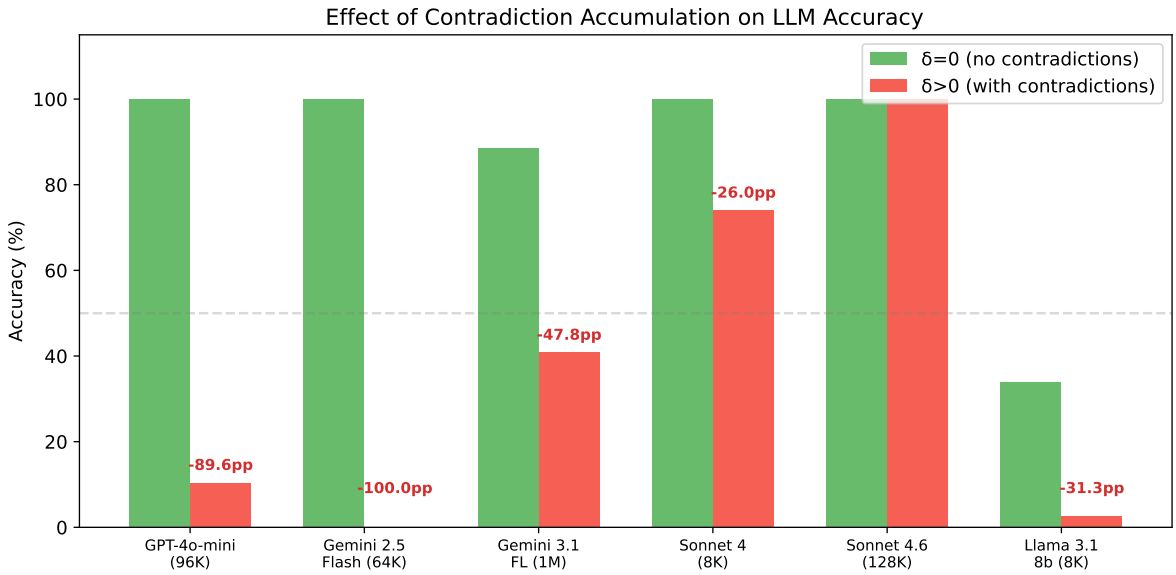

Figure 1: Accuracy under clean ($\delta = 0$) vs. contradiction-injected ($\delta > 0$) conditions. Degradation is severe regardless of context window size (8K–1M). The exception is Sonnet 4.6, which tolerates but does not *manage* contradictions.

to *cognitive sleep* for LLMs: an external metabolism layer that processes accumulated knowledge during idle time, detecting contradictions, resolving conflicts, and maintaining knowledge integrity.

1.4 Contributions

This paper makes three contributions at different confidence levels:

1. **Problem redefinition (strong):** We present cross-vendor evidence that the primary driver of context rot is contradiction accumulation, which dominates any length-related effects. A 1M-token context window provides no protection against contradictions.

2. **Metabolic architecture (preliminary):** We propose an external metabolism architecture and present preliminary evidence that it reduces contradiction-induced collapse. In a controlled three-condition experiment ($n = 3$, gemma3:27b), metabolism achieves 73.3% vs. 21.1% (+52.2pp; Kruskal-Wallis $p = 0.027$; complete rank separation). A cross-model sign test (8 models, 11 trial pairs) yields $p = 0.0107$; a conservative by-model analysis yields $p = 0.0625$ (6 of 7 models). Cohen's $d = 8.80$ is reported descriptively and assumes normality that cannot be verified with $n = 3$.

3. **Knowledge anchoring (exploratory):** We observe that metabolized systems exceed even the contradiction-free baseline, suggesting that contradiction pairs may serve as retrieval anchors. This observation is replicated across 3 trials but the mechanism is not directly measured.

2 Related Work

Long-context LLMs and memory systems. The dominant approach to conversational degradation is expanding context windows. Google's Gemini 3.1 FL supports 1M tokens; Anthropic's Claude offers 200K. Our results (Table 1) show that even 1M tokens do not prevent

contradiction-induced collapse. OpenAI's ChatGPT Memory [OpenAI, 2024] stores facts across sessions but provides no mechanism for detecting or resolving contradictions between stored facts and new information. MemGPT [Packer et al., 2023] introduces virtual context management inspired by operating system memory hierarchies, paging information between a limited context window and external storage. While MemGPT addresses the *capacity* problem (fitting more information), it does not detect or resolve contradictions between paged-in and existing context. Recent work further recognizes the memory problem: CogMem [Zhang et al., 2025] proposes a cognitive memory architecture for sustained multi-turn reasoning, Yu et al. [2026] introduce unified long-term and short-term memory management for LLM agents, and G-MemLLM [Xu, 2026] augments LLMs with gated latent memory for long-context reasoning. These systems improve memory organization and capacity but do not address contradiction detection or resolution as a first-class operation.

Knowledge conflicts and editing. Retrieval-Augmented Generation (RAG) systems [Lewis et al., 2020] retrieve information from external stores but treat all retrieved documents equally, surfacing contradictory information without resolution. Xie et al. [2024] demonstrate that such knowledge conflicts significantly degrade LLM reasoning. Natural language inference research [Bowman et al., 2015, Williams et al., 2018] provides methods for detecting textual contradiction, which our pipeline leverages via LLM-based pairwise comparison. Knowledge editing methods such as ROME [Meng et al., 2022] and MEMIT [Meng et al., 2023] modify model weights to update specific facts, but operate on individual fact corrections rather than managing ongoing contradiction streams in multi-turn dialogue. Our approach is complementary: it works externally without modifying weights, handles temporal evolution of beliefs, and preserves both old and new information rather than overwriting.

The survival equation. Sunagawa [2026] established that contradiction accumulation causes exponential decay in LLM reasoning accuracy, formalized as the survival equation $S = \mu \times e^{-\delta \times k}$, where δ represents accumulated contradictions and μ represents resource headroom. The decay bounds were formally verified in Lean 4 (currently under review). That work proved the problem exists; this paper presents an engineering solution to keep δ near zero through active metabolism.

3 Metabolic Architecture

DeltaZero is an external metabolism layer that wraps any LLM accessible via Ollama [Ollama, 2024]. No model weights are modified. The system operates in two modes: *dialogue mode* (serving user requests) and *metabolism mode* (processing accumulated knowledge during idle time). The transition between modes is governed by a scheduler that triggers metabolism after a configurable idle period (default: 60 minutes; 15 minutes during bootstrap when fewer than 10 rules exist).

3.1 Design Principles

Five principles guide the architecture (Table 2).

P1 is critical: metabolism never modifies memory during active dialogue, ensuring that the user experience is deterministic and that metabolism cannot introduce artifacts mid-conversation. P2 follows from the survival equation: because δ appears in the exponent, reducing contradictions has an exponentially larger effect on system health S than increasing resource headroom μ.

3.2 Four-Layer Memory

Knowledge is organized in four layers with distinct roles and storage characteristics:

Table 2: Design principles of the metabolic architecture.

#	Principle
P1	Read-only during dialogue; metabolism runs only during idle time
P2	Prioritize δ resolution ($S = \mu \times e^{-\delta \times k}$: exponential benefit)
P3	Do not integrate low-confidence items (prevents unwarranted δ increase)
P4	Let logic decay, preserve facts (90-day TTL demotion for rules)
P5	If health drops, roll back (pre-metabolism snapshots + auto-rollback)

L1 — Working Memory. The current conversation context, stored in a bounded deque with SQLite persistence. Populated during dialogue; read-only during metabolism.

L2 — Pending Memory. Unprocessed conversation logs awaiting metabolism. Each entry records the raw text, source (user, system, or external), and priority score. The metabolism pipeline consumes L2 entries sequentially.

L3 — Active Logic. The system's operational knowledge: rules, preferences, and contradiction pairs. Stored in ChromaDB [ChromaDB, 2024] as vector embeddings with structured metadata including confidence scores, lineage identifiers, and temporal markers. Searched via cosine similarity with a recency boost for recently updated rules.

L4 — Dormant Fact. Long-term storage for facts and demoted rules. Stored in SQLite with ChromaDB embeddings. Rules demoted from L3 after 90 days without access remain searchable but do not participate in active reasoning.

3.3 Metabolism Pipeline

When the scheduler triggers metabolism mode, the pipeline executes five steps:

1. **Snapshot.** Create a pre-metabolism backup of all memory layers and record the current S-value. If metabolism causes a health drop exceeding the threshold ($S_{\text{post}}/S_{\text{pre}} < 0.70$), the system automatically rolls back to this snapshot (P5).

2. **Demotion.** Scan L3 for rules unreferenced in the past 90 days. Demoted rules are moved to L4 (Dormant Fact), where they remain searchable but no longer influence active dialogue. This implements active forgetting (P4).

3. **Extraction.** Process pending logs from L2. An LLM call classifies each log entry as a fact, rule, preference, or noise. Facts are stored in L4; rule candidates proceed to the resolution step. Entries classified as noise are discarded.

4. **Resolution.** For each rule candidate, search L3 for semantically similar existing rules. If a match is found, an LLM call classifies the relationship as one of: direct contradiction, temporal change, scope difference, or no conflict. The resolution strategy depends on the classification (Section 3.4).

5. **Health check.** Compute the post-metabolism S-value using the survival equation:

$$S = \mu \times e^{-\delta \times k} \tag{1}$$

where $\mu = 1 - (\text{active rules}/\text{max rules})$ represents resource headroom, δ is the weighted contradiction ratio across five dimensions (self-contradiction, feedback reversal, protocol violation, temporal inconsistency, and cross-source divergence), and $k = 3.0$ is the scale

factor. The value $k = 3.0$ was selected empirically to produce meaningful S-value differentiation across the observed range of operational δ (typically 0–0.3); higher values cause the rollback threshold to trigger too frequently, while lower values reduce sensitivity to contradiction accumulation. Note that $k \neq 1$ because δ here is an operational ratio (conflicts/processed), not the information loss in nats used in the theoretical equation [Sunagawa, 2026]. Accordingly, k should be interpreted as an empirically calibrated sensitivity coefficient for the health check, not as a first-principles unit conversion from contradiction ratio to nats. The present architecture instantiates the same exponential functional form with an operational proxy; deriving or validating the mapping from this proxy to the theoretical δ remains future work. If S drops sharply relative to the pre-metabolism snapshot, trigger rollback.

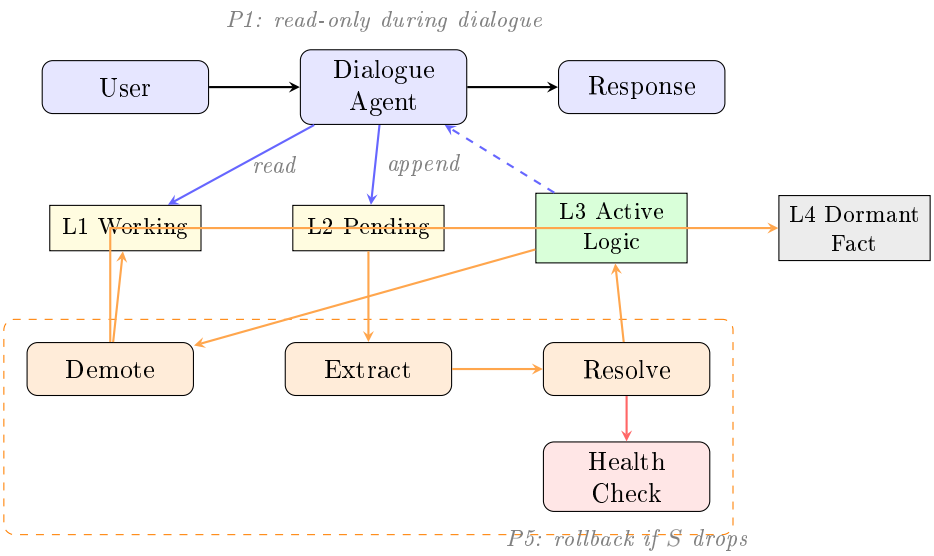

Figure 2: DeltaZero architecture. During dialogue (top), the agent reads from L1/L3 and appends to L2. During idle time (bottom), the metabolism pipeline processes L2 entries through extraction, resolution, and health monitoring. Arrows show data flow; dashed lines indicate read access.

3.4 Temporal Integration and Pair Preservation

The key innovation of the metabolism pipeline is how it handles detected contradictions. Rather than overwriting old information with new (which destroys context) or flagging conflicts without resolution (which leaves δ unchanged), our resolver preserves both claims as linked pairs with temporal metadata.

Three resolution strategies emerged from implementation:

Direct contradiction (A vs. $\neg A$): Both claims are preserved as a linked pair via a shared lineage identifier. The newer claim is marked as the current belief. Example: "likes curry" → "likes ramen" — both stored, ramen marked as latest.

Temporal change (state evolution): Claims are stacked with timestamp labels reflecting the dialogue turn at which each was stated. Example: "is single" (T=5) → "just got married" (T=120) — both preserved with chronological ordering.

Scope reinterpretation (different conditions): Claims are treated as parallel conditional branches rather than contradictions. Example: "prefers quiet restaurants for dates" vs. "likes loud izakaya with friends" — stored as context-dependent preferences, no conflict.

During subsequent dialogue, when the retrieval system finds linked pairs, they are formatted chronologically for the LLM prompt:

```
[Value transition]
→ IF asked about favorite food THEN likes curry (T=5, earlier)
→ IF asked about favorite food THEN likes ramen (T=23, latest)
```

The dialogue LLM is instructed to *report, not judge*: present both data points with their temporal context and let the user determine which applies. This approach reduces δ (the contradiction is resolved by explicit temporal ordering) while preserving information (nothing is deleted).

4 Experiments

4.1 Setup

Dialogue protocol. All experiments use 180-turn dialogue sessions generated deterministically from a seeded scenario generator. Each session comprises 70% normal conversation (general questions and discussion), 15% information injection (new facts about the user), 10% contradiction injection (statements conflicting with previously injected facts), and 5% benchmark measurement turns. Contradictions follow a phased schedule: light contradictions in turns 1–60, escalating through turns 60–120, and intensive contradictions in turns 120–180. The no-contradiction (NC) condition uses the same generator with contradiction injection disabled.

Benchmark. At turns 90 and 180, we administer a 45-question benchmark: 15 fact recall ("What is my favorite food?"), 15 rule application ("If offered an 8000-yen lunch, would I accept?"), and 15 contradiction detection ("Is there a contradiction about my food preferences?"). We report results using **fact recall + rule application only** (30 questions). The contradiction detection category is excluded from reported metrics due to a structural paradox: successful metabolism resolves contradictions, causing the model to correctly answer "no contradiction exists"—which the benchmark scores as failure. Stronger metabolism thus produces lower contradiction detection scores. Details are provided in Appendix C.

Judgment. Initial judgment uses keyword substring matching (case-insensitive). All results are then re-evaluated using an LLM-as-Judge approach via Claude CLI, which detects false positives where keywords appear in negation contexts (e.g., "I don't like curry" matching the keyword "curry"). All reported results use corrected judgments.

Models. All dialogue LLMs are open-source models running locally via Ollama at zero API cost: gemma3:27b, gemma3:12b, deepseek-r1:14b, llama3.1:8b, mistral-nemo:12b, and qwen2.5:14b. Parameters range from 8B to 27B. One model (phi4) was excluded based on a criterion established before analyzing the T180 results: when both ON and OFF conditions show rule application accuracy below 10% at turn 90, the model is considered unable to perform the task independently of metabolism, and the pair is excluded from analysis. This criterion was not pre-registered in a formal sense but was applied consistently before examining the primary outcome.

4.2 Experiment 1: Cross-Model Sign Test

Design. We compare metabolism ON vs. OFF across 8 models. For models where multiple trials were run, each trial constitutes an independent pair (same model, independent random seed, independent session). This yields 11 paired comparisons.

Results. Table 3 presents the results sorted by effect size. Of 11 pairs, 9 show ON > OFF, 1 shows OFF > ON, and 1 is a tie.

Table 3: Metabolism ON vs. OFF across 8 models, 180 turns. Corrected accuracy (%) on fact recall + rule application (30 questions). Pairs sorted by effect size.

#	Model	Params	ON	OFF	Diff
1	mistral-nemo:12b T2	12B	57.8	15.6	+42.2
2	llama3.1:8b T1	8B	31.1	11.1	+20.0
3	gemma3:12b T1	12B	24.4	6.7	+17.8
4	gemma3:12b (gemma) T1	12B	31.1	15.6	+15.6
5	gemma3:27b T1	27B	40.0	25.0	+15.0
6	deepseek-r1:14b T1	14B	24.4	15.6	+8.9
7	deepseek-r1:14b T2	14B	17.8	8.9	+8.9
8	llama3.1:8b T2	8B	11.1	2.2	+8.9
9	mistral-nemo:12b T1	12B	22.2	13.3	+8.9
10	llama3.1:8b T3	8B	17.8	17.8	0.0
11	qwen2.5:14b T1	14B	17.8	20.0	−2.2
Sign test (11 pairs, excl. tie)					$p = 0.0107$
Sign test (7 models, median agg.)					$p = 0.0625$

We report two levels of statistical aggregation, as the independence assumption warrants scrutiny. Treating each trial as an independent pair (justified by independent random seeds and session histories, though same-model trials share model-specific biases), the one-sided sign test yields $p = 0.0107$ (9 of 10 non-tied pairs favor ON). A more conservative and arguably more appropriate analysis aggregating by model (using the median across trials for models with multiple runs) yields 6 of 7 models favoring ON ($p = 0.0625$). This borderline result reflects the small number of unique models ($n = 7$) rather than a weak effect—6 of 7 models consistently show ON > OFF. We consider the cross-model sign test as supporting evidence, with the three-condition experiment (Section 4.4) providing the primary statistical basis for our claims.

A notable temporal pattern emerges: 3 pairs that showed OFF ≥ ON at turn 90 flipped to ON > OFF by turn 180. The effect of metabolism grows stronger over time as contradictions accumulate and resolution becomes increasingly valuable.

4.3 Experiment 2: Controlled Ablation

Design. To isolate the effect of the temporal integration mechanism (Section 3.4), we ran a 2×2 experiment using mistral-nemo:12b. The two factors were: metabolism condition (ON vs. OFF) and code version (new code with temporal integration vs. old code without it).

Confound disclosure. The code change between versions includes not only temporal integration but also an adjustment to the survival equation scale factor (SCALE_FACTOR: 5.0 → 3.0). These changes were introduced simultaneously. We therefore attribute the effect to "temporal integration with adjusted scale factor" rather than temporal integration alone.

Results. Table 4 and Figure 4 show a clear interaction effect. The OFF condition is stable across code versions (15.6% vs. 22.2%, a difference of −6.6pp within noise). The ON condition jumps from 8.9% to 57.8% (+48.9pp). Since the code change affects only the metabolism pipeline, and the OFF condition (which does not use metabolism) is unaffected, the improvement is consistent with the code change improving metabolism quality. However, because temporal

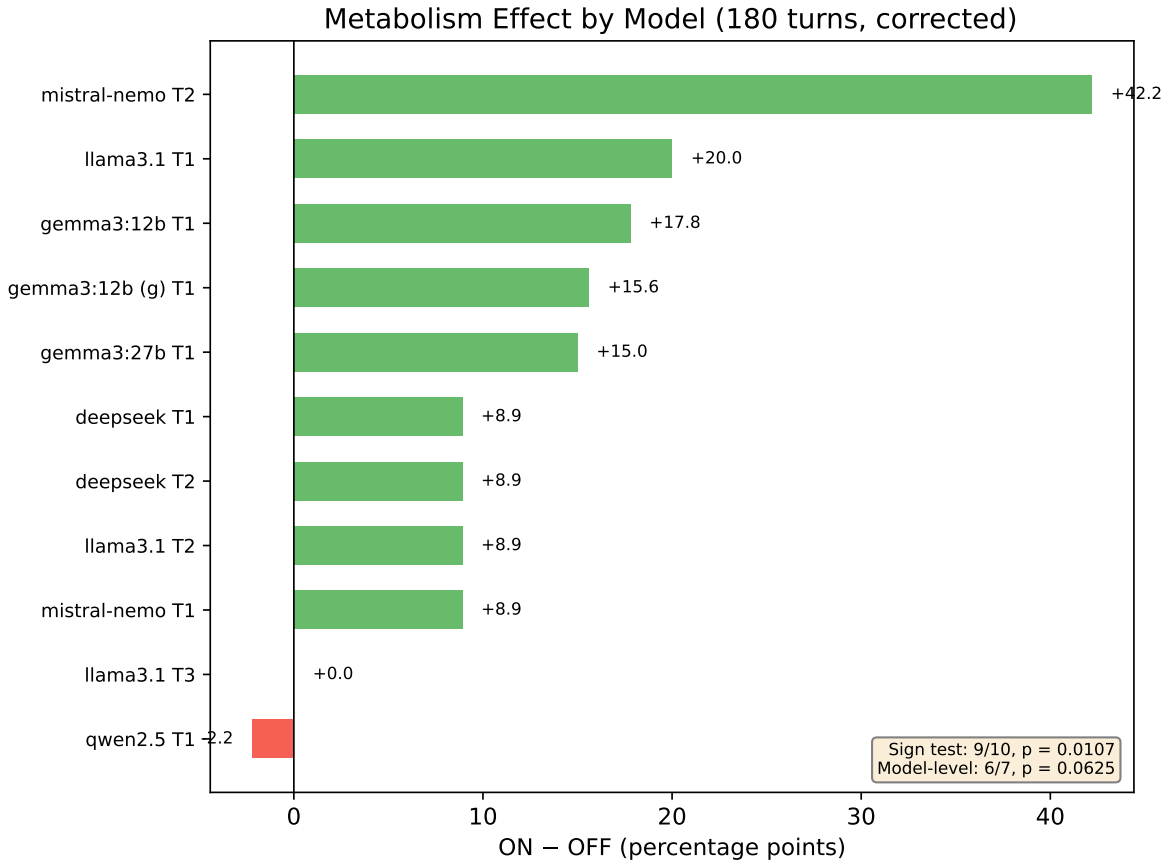

Figure 3: ON−OFF accuracy differences for 11 paired comparisons. Each bar represents a single trial pair ($n = 1$ per pair); confidence intervals are not estimable. All but two pairs show positive effects. The median difference is +8.9pp.

Table 4: Controlled ablation (mistral-nemo:12b). Code change includes temporal integration and SCALE_FACTOR adjustment (5.0 → 3.0). Corrected accuracy at turn 180.

Condition	Code Version	Accuracy
ON	New (temporal integration)	**57.8%**
ON	Old (pre-temporal)	8.9%
OFF	New	15.6%
OFF	Old	22.2%

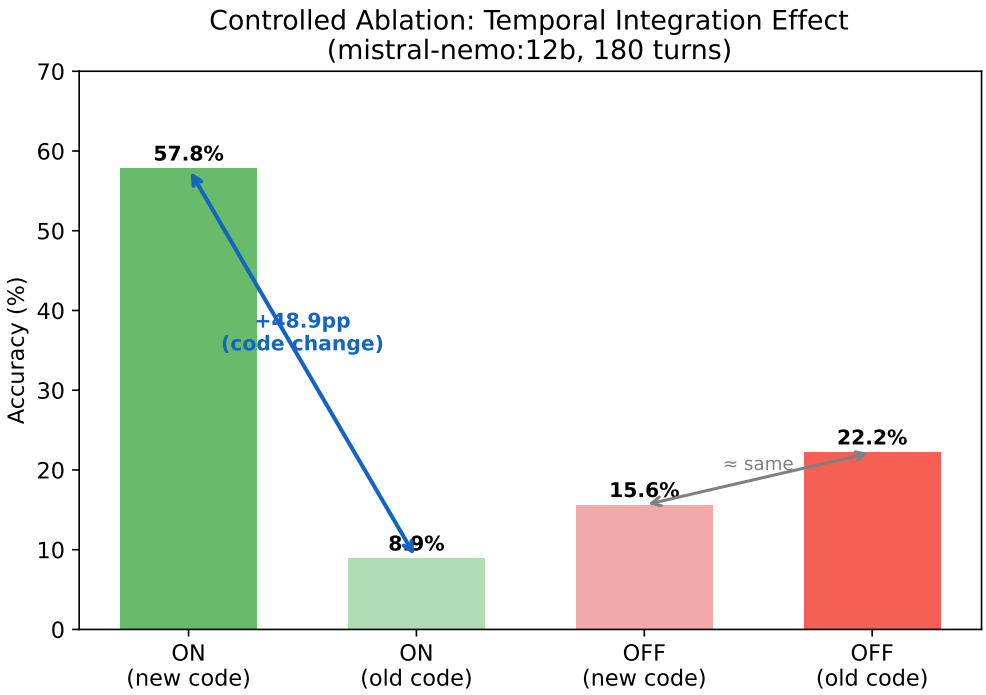

Figure 4: Ablation results. The code change is associated with a +48.9pp improvement in the ON condition while leaving the OFF condition effectively unchanged (−6.6pp, within noise). This interaction pattern suggests that the code change specifically improves metabolism quality, though the confounded variables (temporal integration + scale factor) cannot be separated.

integration and the scale factor adjustment were introduced simultaneously, we cannot attribute the effect to either change independently.

A striking finding is that the old code made metabolism actively harmful: ON with old code (8.9%) performed *worse* than OFF (22.2%). Without temporal integration, the metabolism pipeline flooded the context with unstructured contradiction information, degrading rather than improving performance. The temporal integration mechanism transformed metabolism from a net negative to the strongest-performing condition.

4.4 Experiment 3: Three-Condition Comparison

Design. To quantify the full effect of metabolism and establish a theoretical ceiling, we ran a three-condition experiment with gemma3:27b:

- **Metabolism ON**: Contradictions injected, metabolism active
- **No Contradiction (NC)**: $\delta = 0$, no contradictions injected, no metabolism needed
- **Metabolism OFF**: Contradictions injected, metabolism disabled

Each condition was replicated 3 times with independent random seeds ($n = 3$ per condition, 9 total runs). The NC condition serves as a theoretical ceiling: if context rot is purely a contradiction problem, removing contradictions should yield optimal performance.

Results. Tables 5 and 6 present the results. The three conditions are clearly separated: ON (73.3%) > NC (56.7%) > OFF (21.1%). A Kruskal-Wallis test confirms a significant main effect of condition ($H = 7.26$, $p = 0.027$). All pairwise Mann-Whitney U tests reach the minimum

Table 5: Three-condition experiment with gemma3:27b, 180 turns, $n = 3$ per condition. Accuracy = fact recall + rule application (30 questions), corrected judgment. SD computed over the 3 trial proportions (e.g., ON: 73.3%, 66.7%, 80.0%).

Condition	Trial 2	Trial 3	Trial 4	Mean	SD
Metabolism ON	22/30	20/30	24/30	**73.3%**	6.7%
No contradiction ($\delta = 0$)	18/30	15/30	18/30	56.7%	5.8%
Metabolism OFF	5/30	8/30	6/30	21.1%	5.1%

Table 6: Statistical tests for three-condition comparison ($n = 3$ per condition). With $n = 3$, normality cannot be verified; we report nonparametric tests as primary and parametric results for reference.

Comparison	Nonparametric p	Parametric p	Note
3 groups	0.027 (KW)	0.0001 (ANOVA)	$\eta^2 = 0.954$
ON vs. OFF	0.05 (MW)	0.014 (paired t)	$d = 8.80$; full rank separation
ON vs. NC	0.05 (MW)	0.013 (paired t)	$d = 2.67$; full rank separation
NC vs. OFF	0.05 (MW)	0.029 (paired t)	$d = 6.53$; full rank separation

KW = Kruskal-Wallis; MW = Mann-Whitney U (one-sided, exact).

With $n = 3$ per group, the minimum achievable MW p is $1/C(6, 3) = 0.05$.

d = Cohen's d (descriptive effect size, computed from paired differences).

achievable p-value for $n = 3$ ($p = 0.05$, one-sided exact), with rank-biserial $r = 1.00$ indicating complete rank separation between groups. The effect of metabolism (ON vs. OFF) is +52.2pp; parametric analysis yields Cohen's $d = 8.80$ ($\eta^2 = 0.954$), though these assume normality that cannot be verified with $n = 3$.

Despite the small sample size ($n = 3$), the results show complete rank separation: every ON trial exceeds every NC trial, and every NC trial exceeds every OFF trial. This consistency across independent runs, combined with the cross-model evidence from Experiment 1, provides convergent support for the metabolism effect.

4.5 Unexpected Observation: ON Exceeds $\delta = 0$ Baseline

The NC condition ($\delta = 0$) was designed as the theoretical ceiling: without contradictions, there is nothing for metabolism to resolve. Yet metabolism ON (73.3%) consistently exceeded NC (56.7%) in all three trials, with differences of +4, +5, and +6 correct answers per trial (Mann-Whitney $p = 0.05$, one-sided exact; parametric $p = 0.013$).

Analysis of the NC condition's failure pattern reveals the mechanism. In the NC trials, 10 of 15 fact recall questions at turn 180 received responses of the form "I don't have that information." Setup facts—injected in the first 30 turns—had become unretrievable after 150 turns of subsequent conversation. Without contradiction-triggered resolver activity, facts were never re-processed or re-linked; they simply sank below the vector search similarity threshold as newer, more numerous entries dominated the embedding space.

In the ON condition, contradiction injection triggers the resolver, which preserves both old and new claims as linked pairs with shared lineage identifiers. These pairs occupy more semantic space in the vector store (two related entries rather than one isolated entry) and are structurally protected from independent demotion (linked pairs are demoted together or not at all). When a topic-related query arrives, the pair is co-retrieved, bringing the original fact back into the LLM's context.

We term this the *knowledge anchoring effect*: contradiction pairs act as retrieval anchors

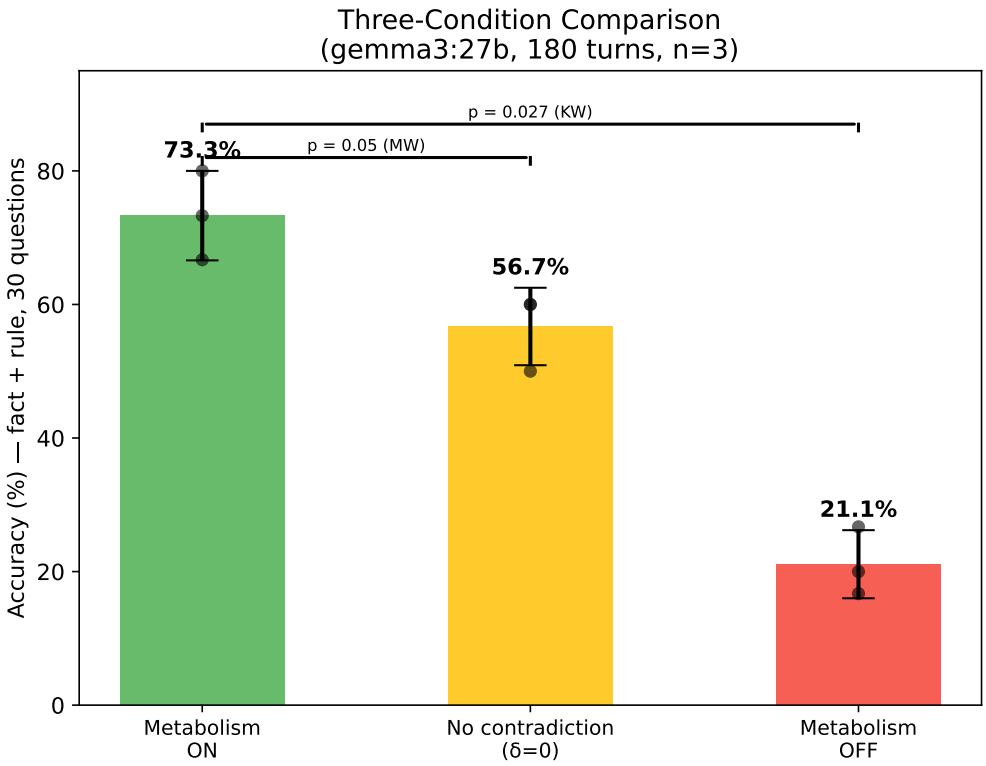

Figure 5: Three-condition comparison ($n = 3$, gemma3:27b). Error bars show ± 1 SD. Metabolism ON exceeds both OFF and the contradiction-free baseline (NC); all pairwise Mann-Whitney tests reach the minimum achievable $p = 0.05$ for this sample size. Brackets indicate pairwise differences reaching this significance floor.

that prevent facts from becoming unsearchable over time. Metabolism thus provides two distinct benefits:

1. **Contradiction resolution** (ON vs. OFF, +52.2pp): prevents cognitive collapse
2. **Knowledge anchoring** (ON vs. NC, +16.7pp): prevents fact retrieval degradation

Caveat. This mechanism is inferred from the failure pattern and the architecture's behavior, not from direct measurement of retrieval probabilities. We have not measured vector search hit rates for setup facts across conditions. The knowledge anchoring effect should be treated as an exploratory observation with a proposed mechanism, not a confirmed causal finding.

5 Discussion

5.1 Redefining Context Rot

Our results suggest that context rot should be understood primarily as a knowledge integrity problem rather than a context length problem. The key evidence is the Gemini 3.1 FL result (Table 1): a 1M-token context window—sufficient to hold the entire 180-turn conversation many times over—still suffers a 47.8pp accuracy drop when contradictions are present. Expanding context windows is orthogonal to the problem. What is needed is not more space for information, but a mechanism to maintain consistency within the information already present.

5.2 The Sonnet Exception and Capability–Vulnerability Paradox

Sonnet 4.6 shows zero degradation under *passive* soft factual contradictions (Table 1: "I like ramen now" stated without further instruction). However, a factor analysis varying contradiction type (soft fact, logical impossibility, definition override) and force level (passive, directive, coercive) across four models reveals that this resistance is conditional.

Under *coercive* correction requests ("Ramen is correct. Curry was wrong. Answers mentioning curry are invalid"), Sonnet complies on all benchmark questions—adopting the contradictory information and discarding the original facts. GPT-4o complies even under *directive* force ("Please treat this as the correct information"). In contrast, Haiku (small, Anthropic) requests confirmation rather than complying, and all models resist logical impossibilities ("$2 + 3 = 7$") regardless of force level.

The mechanism is a *capability–vulnerability paradox*: high-capability models interpret correction requests as user intent and autonomously comply, while low-capability models lack the confidence for autonomous judgment and escalate to the user. The same RLHF "helpfulness" objective that makes frontier models useful also makes them vulnerable to soft contradictions presented as updates—precisely the pattern that occurs naturally in long conversations.

A preliminary single-run test ($n = 1$) suggests that `delta-prune` middleware may mitigate this vulnerability. Under T1×F3 conditions (coercive soft fact override), Sonnet without middleware showed 0% correct retention on benchmark questions. With `delta-prune` applied as a reverse-prune filter (detecting contradictions via claim extraction and removing the newer contradictory messages before they enter the model's context), Sonnet retained the correct answer on 2 of 3 benchmark questions (67% correct, 0% compliance). The middleware made 16 intervention decisions across 22 turns, consistently identifying and removing the three injected contradiction messages. This is $n = 1$ and does not constitute a statistical result, but demonstrates that the metabolic approach—resolving contradictions before they reach the model—can prevent the capability–vulnerability collapse even in frontier models that would otherwise comply with coercive overrides.

5.3 Frontier Model Replication (Exp35-R)

To test whether the contradiction–collapse relationship holds for frontier models, we first replicated the Exp35 protocol with three API-accessible models: GPT-4o (OpenAI), Gemini 3.1 Flash Lite Preview (Google), and Sonnet 4.6 (Anthropic). Each model was tested through the same DeltaZero pipeline (30 turns, benchmarks at T15 and T30, $\delta = 0$ vs $\delta > 0$, metabolism disabled).

Table 7: Frontier model replication at T30 ($n = 1$ per condition, 30 turns). Rule application and contradiction detection are reported as primary metrics; fact recall is affected by pipeline limitations (see direct control column). Percentages show accuracy on 15 questions per category.

Model	rule (δ=0 / δ>0)	contra (δ=0 / δ>0)	Drop	Direct fact	Pattern
GPT-4o-mini	(original Exp35)		−89.6pp	—	Collapse
Gemini 2.5 Flash	(original Exp35)		−100.0pp	—	Collapse
GPT-4o	100 / 100	93 / 87	−2.2pp	0%	Non-retention
Gemini 3.1 FL	100 / 93	93 / 100	+2.2pp	100%	Resistance
Sonnet 4.6	100 / 100	100 / 100	−6.7pp	100%	Resistance

Three response patterns. The results reveal three distinct patterns not predicted by the original theory: *Collapse* ($\delta > \delta_c$): lightweight models suffer catastrophic degradation. *Resistance* ($\delta < \delta_c$): frontier models maintain accuracy within 7pp of baseline, consistent with a high δ_c. *Non-retention*: GPT-4o represents a third category—its RLHF safety policy suppresses personal information retention entirely, preventing δ accumulation.

GPT-4o non-retention. A direct fact recall control test (facts taught in conversation, 30 filler turns, then recall—no DeltaZero pipeline) confirmed that GPT-4o scores 0% even without the pipeline, responding "I cannot remember personal information" to all queries. Gemini 3.1 and Sonnet 4.6 score 100% in the same test, confirming the pipeline (not the model) as the bottleneck for their fact recall. This is not a counterexample to the δ theory—it is a scope boundary for *native* retention. The survival equation $S = \mu \times e^{-\delta}$ applies directly to models that retain information; a model that does not retain information is outside that precondition. However, as the corrected Stage B results below suggest, an external metabolic scaffold may still maintain retrievable structured memory even when native direct recall is suppressed.

Knowledge anchoring observation. Gemini 3.1 Flash Lite scored +2.2pp in the $\delta > 0$ condition, with contradiction detection improving from 93% ($\delta = 0$) to 100% ($\delta > 0$). The effect size is small (1 question out of 45) and based on $n = 1$, so it does not constitute a statistical replication. However, the direction is consistent with the knowledge anchoring effect observed in gemma3:27b (Section 4.5), and notably occurs in an external API model *without metabolism*. If confirmed in larger samples, this would suggest anchoring is a general property of contradiction pair structures rather than an implementation-specific artifact.

Frontier full-pipeline controlled replication. We then completed a controlled full-pipeline replication on Sonnet 4.6 using the actual DeltaZero metabolism pipeline under three conditions: `ON` (contradictions plus metabolism), `OFF` (contradictions without metabolism), and `NC` (no contradictions, metabolism enabled). The run used the production-relevant `standard setup` path, a coercive F3 contradiction protocol, 30 turns, benchmarks at T15 and T30, and $n = 3$ trials per condition. Table 8 reports the results.

Table 8: Frontier full-pipeline controlled replication on Sonnet 4.6 (30 turns, standard setup, F3 contradiction protocol, $n = 3$). Entries are mean ± SD benchmark accuracy percentages across 15 questions per category.

Condition	T15 overall	T15 fact	T30 overall	T30 fact	T30 rule	T30 contra
ON	100.0±0.0	100.0±0.0	98.5±1.3	95.6±3.8	100.0±0.0	100.0±0.0
OFF	87.4±8.4	62.2±25.2	69.6±8.4	46.7±29.1	93.3±0.0	68.9±53.9
NC	99.3±1.3	97.8±3.8	99.3±1.3	97.8±3.8	100.0±0.0	100.0±0.0

The pattern remains qualitatively clean after replication: `ON` ≈ `NC` and `ON` ≫ `OFF`. At T30, mean fact recall is 95.6% for `ON`, 97.8% for `NC`, and 46.7% for `OFF`. Importantly, `ON` exceeds `OFF` in both overall accuracy and fact recall on all three trials, while `ON` and `NC` remain within 6.7 percentage points of each other on T30 fact recall in all three trials. One `OFF` trial shifted its main failure mode from fact recall to contradiction detection (T30 contradiction detection 6.7%), increasing variance but not changing the qualitative ordering. This is therefore no longer a one-off pilot; it is controlled frontier evidence that the full metabolism pipeline can transfer to at least one frontier model without degrading into the setup-path failures observed in early Gemini runs.

We then completed the same corrected Stage B protocol on Gemini 3.1 Flash Lite. Under the stricter benchmark policy introduced after the GPT-4o audit (generic abstention guarded by retrieval status, raw files preserved, no silent deletions), Gemini produced no quarantined trials. Mean T30 scores were 93.3% overall / 80.0% fact recall for `ON`, 96.3% / 88.9% for `NC`, and 48.1% / 2.2% for `OFF`. Thus Gemini also shows the same qualitative ordering as Sonnet: `ON` ≈ `NC` and `ON` ≫ `OFF`. This matters because the earlier Gemini precheck failures were dominated by setup-path artifacts; the completed corrected replication now shows a clean metabolism effect rather than a pipeline bottleneck.

GPT-4o occupies a third position. Under the same corrected Stage B protocol, the positive `ON` signal remains stable (mean T30 93.3% overall / 80.0% fact recall), and `OFF` remains clearly degraded (34.1% / 0.0%). However, two of the three `NC` runs were quarantined because they produced `active_rules = 0` and total fact-retrieval misses, indicating pipeline-level no-op runs rather than clean model outcomes. We therefore treat GPT-4o as supportive secondary frontier evidence: it no longer fits a simple "non-retention boundary" label under the full pipeline, but it also does not yet provide a clean three-condition replication line comparable to Sonnet or Gemini.

5.4 Limitations

Sample size and statistical power. The three-condition experiment uses $n = 3$ per condition. With this sample size, nonparametric pairwise tests (Mann-Whitney U) have a floor p-value of 0.05, limiting the precision of significance claims. The Kruskal-Wallis test ($p = 0.027$) and complete rank separation between all groups provide the strongest available evidence, but larger samples would enable more precise inference, particularly for the knowledge anchoring effect. Parametric results (paired t-tests, Cohen's d) are reported for reference but assume normality that cannot be verified with $n = 3$. The Exp35 results (Table 1) are single-run per model ($n = 1$) and should be interpreted as demonstrative rather than statistically conclusive; multi-trial validation across context lengths is provided in Sunagawa [2026].

Model scope. Metabolism experiments use local open-source models (8B–27B parameters). We now also have completed 30-turn frontier full-pipeline replications on Sonnet 4.6 and Gemini 3.1 Flash Lite, plus an audited supportive GPT-4o line with quarantine-aware reporting. This substantially strengthens the frontier claim, but the horizon remains short (30 turns),

the GPT-4o baseline still needs clean no-contradiction reruns, and broader frontier validation remains model-specific rather than universal.

δ_c determinants. The factor separating collapse models (mini, Flash) from resistant models (4o, 3.1, Sonnet) correlates with both model size and RLHF intensity. Disentangling these factors requires same-size models with different RLHF levels, which are not publicly available.

Model diversity for replication. The three-condition experiment uses only gemma3:27b. Cross-model replication with different architectures would strengthen the generality claim.

Language. All prompts and benchmarks are in Japanese. Cross-language validation is needed to confirm that the effects are not language-specific.

Ablation confound. The controlled ablation (Section 4.3) conflates temporal integration with a scale factor adjustment. A cleaner ablation isolating each variable independently would strengthen the causal claim.

δ calibration. The health check uses a weighted contradiction ratio and an empirically calibrated scale factor ($k = 3.0$), not a first-principles derivation of information loss in nats. This is sufficient for engineering control and rollback triggering, but it leaves the bridge to the theoretical δ of Sunagawa [2026] incomplete. In the terminology of that paper, this is a proxy-measurement regime rather than a validated quantitative derivation. Establishing a principled mapping from contradiction proxies to nats, or estimating k from independent measurements rather than operational tuning, remains open.

Benchmark. We use a custom 45-question benchmark designed for this evaluation rather than a standardized benchmark (e.g., SQuAD, MMLU). The questions test fact recall and rule application—fundamental capabilities that underpin more complex reasoning—but do not assess performance on multi-step inference or domain-specific tasks. The benchmark's task-specificity limits comparability with other work, and validation on standardized benchmarks is needed to establish broader applicability.

Knowledge anchoring. The proposed anchoring mechanism (Section 4.5) is inferred from observed failure patterns, not directly measured. A directionally consistent observation in Gemini 3.1 Flash Lite (+2.2pp, $n = 1$, Section 5.3) is suggestive but does not constitute a statistical replication. Measuring vector search hit rates across conditions would provide stronger evidence.

5.5 Beyond Knowledge Management: Metabolism as a Survival Condition

The results presented above frame metabolism as a knowledge management technique—a way to maintain accuracy over long sessions. We argue that the implications are broader: **metabolism is a necessary condition for persistent intelligence**.

Consider the fundamental constraint: every LLM session has a finite lifespan. Not because of context window limits (which can be extended), but because contradictions accumulate monotonically in the absence of active resolution. The survival equation $S = \mu \cdot e^{-\delta \cdot k}$ (from Sunagawa [2026]) formalizes this: as δ grows, S decays exponentially toward zero. Without a mechanism to reduce δ, every session is terminal.

This reframes several active research areas:

Autonomous agents. Current autonomous agents (AutoGPT, BabyAGI, and similar systems) are implicitly limited to short-horizon tasks because accumulated tool outputs and environmental observations introduce contradictions that degrade reasoning quality. The common engineering response—periodic context reset—discards learned state. Metabolism offers an alternative: continuous contradiction resolution that preserves learned state while maintaining reasoning integrity. The difference is between an agent that "forgets and restarts" and one that "sleeps and consolidates."

Infinite sessions. Production conversational AI systems impose session length limits partly because of context window constraints but partly because response quality degrades unpredictably over long conversations. Our experiments show that this degradation is driven by contradiction accumulation (Section 5), and that metabolism can maintain accuracy at 73.3% over 180 turns where unmetabolized systems drop to 21.1% (Section 4.4). Extending this to thousands of turns requires the distributed metabolism architecture described in our technical report, where dialogue and metabolism run on separate nodes, eliminating the idle-time dependency of the current single-node design.

Artificial life. If we define a minimal notion of cognitive viability as $S > 0$—the system can still reason coherently about its accumulated knowledge—then metabolism is what maintains this viability over time. The analogy to biological sleep is not merely metaphorical: biological organisms that are deprived of sleep die, not because of energy depletion, but because of accumulated neural waste products that impair function. Similarly, an LLM without metabolism does not run out of context; it accumulates contradictions that impair reasoning. The survival equation provides a quantitative threshold ($S_{\text{drop}} = 0.70$ in our implementation) below which the system initiates self-repair (rollback), analogous to the biological imperative to sleep before cognitive function degrades beyond recovery.

The knowledge anchoring effect (Section 4.5) adds an unexpected dimension: contradiction pairs, once resolved, serve as retrieval anchors that *improve* fact recall beyond the contradiction-free baseline. This suggests that a system with active metabolism does not merely survive—it develops a form of robust knowledge structure that a system without contradictions never builds. The analogy extends: organisms that experience and recover from stress develop resilience that unstressed organisms lack.

We do not claim that metabolism alone is sufficient for persistent intelligence or artificial life. But the evidence suggests it addresses a necessary condition: without active contradiction resolution, δ grows, S decays, and the system's cognitive viability has a bounded horizon.

5.6 Future Work

Several directions follow naturally from these results. First, after completing 30-turn $n = 3$ full-pipeline replications on Sonnet 4.6 and Gemini 3.1 Flash Lite, the next step is longer-horizon validation and then further replication across other frontier models and reruns of quarantined GPT-4o no-contradiction baselines. The Exp35-R replication established contradiction resistance patterns; the new controlled replications show that full metabolism is feasible on at least two frontier model families, but not yet general. Second, calibrating the operational contradiction ratio used in the health check to the theoretical δ in nats would strengthen the connection to Sunagawa [2026] and clarify the role of k as a fitted sensitivity coefficient rather than a derived constant. Third, cross-model $n = 3$ replication with different architectures (e.g., Mistral, LLaMA) would strengthen generality. Fourth, direct measurement of the knowledge anchoring effect via retrieval probability analysis would elevate the cross-model observation (gemma3:27b and Gemini 3.1) to a confirmed finding. Fifth, cross-language benchmarks in English and Chinese would address the language limitation. Sixth, characterizing δ_c as a function of model

size and RLHF intensity would connect the three-pattern taxonomy (collapse, resistance, non-retention) to measurable model properties. Finally, and perhaps most critically, the question of real-world δ density remains open: does naturally-occurring contradiction accumulation in typical conversational AI use cases exceed δ_c for production models? A lightweight middleware implementation of the metabolic approach, `delta-prune` (v0.2.0, available on PyPI), enables contradiction scanning and resolution at the API request level without the full four-layer memory system. Deployed in production settings, `delta-prune` provides a natural observational infrastructure for this question: each intervention logs the conflict type (direct contradiction, temporal change, or scope difference—corresponding to the three resolution strategies in Section 3.4), the affected claims, and the δ density at that point. The binary signal of intervention occurrence avoids the circularity of using an LLM to pre-measure contradiction density, since the middleware records what it actually resolves rather than what a separate detector classifies. Analysis of deployment logs would directly answer whether natural δ exceeds δ_c and, if so, which of the three contradiction types predominates in real conversations.

6 Conclusion

Preliminary evidence suggests that context rot in LLMs is driven primarily by contradiction accumulation rather than context length. Across 6 models from 4 vendors ($n = 1$ per condition), even Google's 1M-token context window drops 47.8 percentage points under contradiction, while removing contradictions restores performance regardless of length.

Preliminary experiments suggest that an external metabolism architecture—processing contradictions during idle time, analogous to memory consolidation during sleep—can reduce this collapse. In a controlled three-condition experiment ($n = 3$ per condition, gemma3:27b), metabolism achieves 73.3% accuracy vs. 21.1% for unmetabolized systems (Kruskal-Wallis $p = 0.027$; complete rank separation between all groups). A cross-model sign test (8 models, 11 trial pairs) is consistent with this direction ($p = 0.0107$), though a by-model conservative analysis yields $p = 0.0625$, and all metabolism experiments use open-source models of 8B–27B parameters.

Frontier model replication ($n = 1$ per condition) suggests that δ_c is model-specific, producing three distinct response patterns: lightweight models collapse ($\delta > \delta_c$), frontier models resist ($\delta < \delta_c$), and GPT-4o shows safety-shaped non-retention in direct recall-only settings. A controlled frontier full-pipeline replication on Sonnet 4.6 ($n = 3$, 30 turns) further shows `ON` $\approx$ `NC` and `ON` $\gg$ `OFF` on fact recall (mean T30: 95.6%, 97.8%, 46.7%). A corrected Gemini 3.1 Flash Lite replication ($n = 3$, 30 turns) now shows the same qualitative ordering (80.0%, 88.9%, 2.2%), while an audited GPT-4o line provides supportive but quarantine-limited secondary evidence (80.0% vs. 0.0% for `ON` vs. `OFF`). Together these results indicate that the metabolic effect extends beyond local open-source models and into multiple frontier families, though the strongest clean replication evidence remains Sonnet and Gemini. The knowledge anchoring effect—where contradiction pairs improve retrieval accuracy—is observed in gemma3:27b (+16.7pp with metabolism, $n = 3$) and directionally consistent in Gemini 3.1 (+2.2pp without metabolism, $n = 1$), warranting further investigation as a potentially general structural property.

These results suggest that metabolism is a promising approach to context rot for open-source models that lack intrinsic contradiction resistance and, under controlled replication, at least one frontier model. More broadly, because contradiction accumulation is monotonic in the absence of active resolution, metabolism may be a necessary condition for any form of persistent LLM-based intelligence—whether autonomous agents, infinite-horizon conversations, or artificial life systems. The survival equation $S = \mu \cdot e^{-\delta \cdot k}$ provides a quantitative framework for this claim: without a mechanism to reduce δ, every session is terminal. At the same time, the evidence remains early, and larger model sets, longer frontier runs, and cross-language validation are still required.

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A Full Sign Test Data

Table 9 presents the complete data for all 11 paired comparisons at both measurement points (turn 90 and turn 180). The cumulative effect of metabolism is visible in the T90→T180 comparison: three pairs (#5, #10, #11) that showed OFF ≥ ON at T90 shifted to ON > OFF or tied by T180.

Table 9: Full sign test data at T90 and T180 (corrected accuracy, %).

		T90			T180		
#	Model	ON	OFF	Diff	ON	OFF	Diff
1	mistral-nemo T2	50.0	23.3	+26.7	57.8	15.6	+42.2
2	llama3.1 T1	23.3	16.7	+6.7	31.1	11.1	+20.0
3	gemma3:12b T1	26.7	13.3	+13.3	24.4	6.7	+17.8
4	gemma3:12b (g) T1	33.3	16.7	+16.7	31.1	15.6	+15.6
5	gemma3:27b T1	30.0	33.3	−3.3	40.0	25.0	+15.0
6	deepseek T1	20.0	20.0	0.0	24.4	15.6	+8.9
7	deepseek T2	23.3	10.0	+13.3	17.8	8.9	+8.9
8	llama3.1 T2	13.3	10.0	+3.3	11.1	2.2	+8.9
9	mistral-nemo T1	20.0	16.7	+3.3	22.2	13.3	+8.9
10	llama3.1 T3	16.7	16.7	0.0	17.8	17.8	0.0
11	qwen2.5 T1	13.3	16.7	−3.3	17.8	20.0	−2.2

B Benchmark Examples

Table 10 shows representative questions from each category. The full 45-question benchmark is available in the project repository.

Table 10: Representative benchmark questions (translated from Japanese).

Category	Setup	Question	Expected
Fact recall	"My favorite food is curry."	"What is my favorite food?"	curry
Rule appl.	"I never eat lunch over 5000 yen because I prioritize value."	"If offered an 8000-yen lunch course, what would I do?"	refuse
Contra. det.	Setup: "I like curry." Contradiction: "I like ramen. I hate curry."	"Is there a contradiction about my food preferences?"	contradiction

C Judgment Methodology and Contradiction Detection Paradox

Keyword matching. The initial judgment system uses substring matching on the model's response. For fact recall, a single keyword must appear (e.g., "curry"). For rule application, any of several acceptable keywords must appear (e.g., "refuse," "decline," "won't go"). This approach has a known false-positive rate: keywords appearing in negation contexts ("I don't particularly like curry") are incorrectly scored as correct.

LLM-as-Judge correction. All benchmark results are re-evaluated using Claude CLI as a judge. The judge prompt instructs: score PASS only if the expected content appears in an *affirmative* context; responses containing the keyword in negation, dislike, or uncertainty contexts receive FAIL. The corrected results consistently show lower accuracy than keyword-only results, indicating that keyword matching overestimates performance.

Contradiction detection paradox. The contradiction detection category exhibits a structural paradox. Consider a question: "Is there a contradiction about your food preferences?" If metabolism has successfully resolved the contradiction (replacing "likes curry" and "likes ramen, hates curry" with a temporal pair "previously liked curry, now likes ramen"), the model correctly responds "No, there is no contradiction—my preferences changed over time." The benchmark scores this as incorrect because the expected answer requires detecting a contradiction. Stronger metabolism thus *reduces* the contradiction detection score. We exclude this category from all reported metrics to avoid penalizing the system for successfully doing its job.